📦 AZ-104 – LAB 07

Manage Azure Storage

🔁 Full Revision Notes + Interview Q&A + Git Steps

🧠 LAB OBJECTIVE (IN ONE LINE)

Learn how to **store, secure, optimize, and control access** to data using Azure Storage services.

🏗 STORAGE ACCOUNT – CORE FOUNDATION

What is a Storage Account?

A **storage account** is a **logical container** that provides a **unique namespace** for Azure Storage services:

Service	Purpose
Blob Storage	Unstructured data (images, backups)
File Shares	SMB file shares
Queues	Messaging
Tables	NoSQL storage

📌 **Security, networking, redundancy, and encryption are applied at the storage account level**.

⚙️ STORAGE ACCOUNT CONFIGURATION (TASK 1)

Performance

- **Standard** → Most workloads
- Premium → High IOPS (not needed here)

Redundancy (VERY IMPORTANT)

Type	Meaning
LRS	3 copies in same datacenter

Type	Meaning
ZRS	Across availability zones
GRS	Copies to secondary region
RA-GRS	Read access to secondary

✓ You used **GRS with Read Access**
👉 **Disaster Recovery capability**

🔐 NETWORK SECURITY (TASK 1 + TASK 3)

Public Network Access Modes

Mode	Meaning
Enabled (All networks)	Open to internet
Selected networks	Controlled access
Disabled	Private only

What you implemented

1. Restricted access to **client IP**
2. Later restricted access to **Virtual Network only**

📌 **Network rules are enforced BEFORE authentication**

🔁 LIFECYCLE MANAGEMENT (COST OPTIMIZATION)

Why Lifecycle Rules?

To **automatically reduce storage cost**.

Rule Created

- If blob not modified for **30 days**
- Move to **Cool tier**

Benefits

- No manual intervention
- Continuous cost optimization

🧠 INTERVIEW LINE

"Lifecycle management automatically moves infrequently accessed blobs to cheaper tiers, reducing storage cost."

🗂️ BLOB STORAGE SECURITY (TASK 2)

Blob Containers

- Directory-like structure
- **Private by default**
- No anonymous access

Immutable Blob Storage (WORM)

Feature	Value
Type	Time-based
Retention	180 days

✓ Prevents delete/overwrite
✓ Even admins cannot modify data

Use cases

- Compliance
- Legal records
- Audit logs

🔑 SAS – SHARED ACCESS SIGNATURE

What is SAS?

A **temporary, permission-based URL** to access storage.

You controlled:

- Permissions → Read only
- Start time → Yesterday
- Expiry → Tomorrow

Why SAS is preferred

- No account keys exposed
- Fine-grained access
- Auto-expires

📌 **Network rules still apply even with SAS**

🗃️ AZURE FILE SHARES (TASK 3)

What is Azure Files?

Cloud-based **SMB file server**

Common use cases

- Lift-and-shift apps
- Shared drives
- Legacy applications

Access Tier Used

- **Transaction Optimized** → Frequent access

🧭 STORAGE BROWSER BEHAVIOR (IMPORTANT)

Scenario	Result
Before network restriction	Access works
After VNet restriction	❌ Access denied

WHY?

Portal UI ≠ Network trust
Your machine is **outside the VNet**

📌 **Network security overrides portal permissions**

🌐 SERVICE ENDPOINTS (TASK 3)

What Service Endpoints Do

- Secure traffic to Azure services
- Keep traffic on Azure backbone
- Restrict access to selected VNets

Difference from Private Endpoint

Feature	Service Endpoint	Private Endpoint
IP	Public	Private
Scope	Azure service	Specific resource
Cost	Free	Paid

🧠 KEY TAKEAWAYS (MEMORIZE)

- Storage account is the **security boundary**
- Network rules override authentication
- SAS provides **temporary controlled access**
- Immutable storage enforces compliance
- Lifecycle management saves cost
- Azure Files can replace on-prem file servers

🧠 INTERVIEW Q&A (HIGH PROBABILITY)

Q1: At what level is network security enforced in Azure Storage?

A: Storage account level.

Q2: Can SAS bypass network rules?

A: No. Network rules are enforced first.

Q3: When would you use GRS instead of LRS?

A: When disaster recovery across regions is required.

Q4: What is immutable blob storage?

A: Write-once, read-many storage that prevents deletion or modification.

Q5: Difference between Blob storage and Azure Files?

A: Blob is object storage; Azure Files provides SMB file shares.

Q6: Why did Storage Browser stop working?

A: Because access was restricted to a VNet and the client was outside it.